4 **(a)** figure shows an interference pattern (which can only occur when waves from both sources are coherent). B1

(b) intensity at P or $x = 0$ is a minimum B1

since point is equidistant from speakers so path difference = 0 B1

intensity minimum if and only if waves from speaks are emitted anti-phase B1

(c) $x = \lambda D / a$
$0.040 = \lambda\ (1.5) / 0.15\ (\Rightarrow \lambda = 4.0 \times 10^{-3}\ \text{m})$ M1

$v = f\lambda$
$f = 340 / 4.0 \times 10^{-3} = 85\,000\ \text{Hz}$ A1

(d) intensity graph is only zero at the 12 cm mark which is farther from L_1 than L_2. M1

Since intensity decreases with distance, speaker L_1 has a higher power. A1